

Analog-to-Digital & Digital-to-Analog Conversion

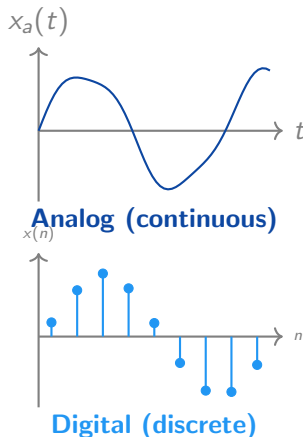
Syem Aziz

Why Do We Need A/D Conversion?

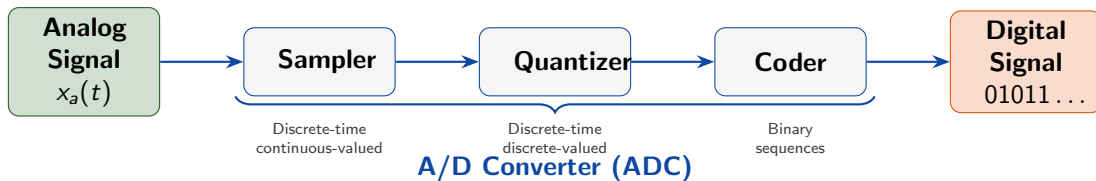
Real-world signals are analog

- Speech, music, ECG, seismic waves
- Radar & sonar signals
- Video & images
- All vary *continuously* in time and amplitude

Problem: Computers only work with **discrete numbers of finite precision.**



The A/D Conversion Pipeline



Key Insight

Although we model ADC as three separate steps, in practice a **single device** performs all three operations. Sampling *always* precedes quantization.

Definition

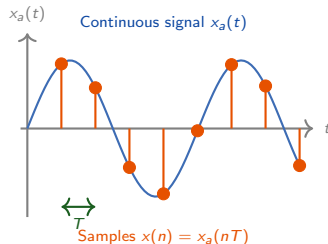
Convert a **continuous-time** signal $x_a(t)$ into a **discrete-time** signal $x(n)$ by measuring its value at equally spaced instants:

$$x_a(nT) = x(n), \quad n \in \mathbb{Z}$$

where T is the **sampling interval** (seconds).

- $f_s = \frac{1}{T}$ is the **sampling frequency** (Hz)
- Sampling is **reversible** — no information is lost if f_s is high enough
- Key condition: **Nyquist–Shannon Sampling Theorem**

$$f_s \geq 2 f_{\max}$$



From Analog Frequency to Digital Frequency

Derivation — Step by Step

Start with a cosine signal:

$$x_a(t) = A \cos(2\pi Ft + \theta)$$

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Since $T = \frac{1}{F_s}$, we get $nT = \frac{n}{F_s}$:

$$x(n) = A \cos\left(2\pi \cdot \frac{F}{F_s} \cdot n + \theta\right)$$

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$$x(n) = A \cos(2\pi f n + \theta) \quad \text{where } f = \frac{F}{F_s}$$

Sampling — Quick Example

Example

A signal has frequency $F = 1000$ Hz. It is sampled at $F_s = 8000$ Hz.

$$f = \frac{F}{F_s} = \frac{1000}{8000} = 0.125 \text{ cycles/sample}$$

The discrete-time signal is:

$$x(n) = A \cos(2\pi \times 0.125 \times n + \theta) = A \cos\left(\frac{\pi}{4} n + \theta\right)$$

Remember

- F (capital) = analog frequency in **Hz**
- f (lowercase) = digital frequency in **cycles/sample** (no unit)
- $f = F/F_s$ links the two worlds

Sampling — The Aliasing Problem

What is Aliasing?

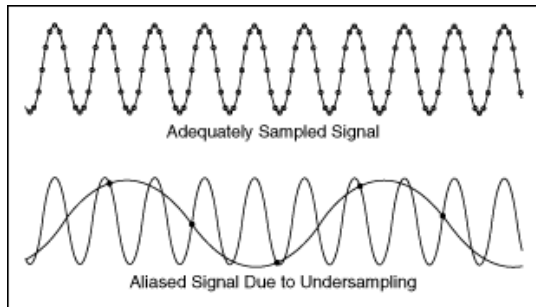
If $f_s < 2f_{\max}$, high-frequency components **masquerade** as lower frequencies in the sampled signal — *this is aliasing*.

How to Avoid It

- Ensure $f_s \geq 2f_{\max}$ (Nyquist rate)
- Use an **anti-aliasing filter** before sampling to band-limit the signal

Example: Audio CD uses $f_s = 44.1$ kHz because human hearing reaches ≈ 20 kHz, so $2 \times 20 = 40$ kHz is the minimum required rate.

The Aliasing Problem



What is Aliasing?

Definition

When $f_s < 2f_{\max}$, high-frequency components **masquerade as lower frequencies** in the sampled signal. This is **aliasing**.

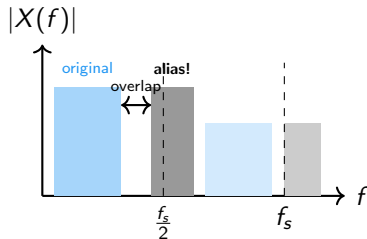
A signal at frequency f sampled at f_s *appears* to be at alias frequency:

$$f_{\text{alias}} = |f - k f_s|, \quad k \in \mathbb{Z}$$

Choose k such that $f_{\text{alias}} \in [0, \frac{f_s}{2}]$.

Example: $f = 900 \text{ Hz}$, $f_s = 1000 \text{ Hz}$

$$f_{\text{alias}} = |900 - 1 \times 1000| = 100 \text{ Hz}$$



The Nyquist–Shannon Sampling Theorem

Theorem (Nyquist–Shannon, 1949)

A band-limited signal with highest frequency $f_{\max}$ can be *perfectly* reconstructed from its samples **if and only if**:

$$f_s \geq 2 f_{\max}$$

The minimum rate $f_N = 2 f_{\max}$ is called the **Nyquist rate**.

In angular frequency:

$$\Omega_s \geq 2 \Omega_{\max}$$

where $\Omega_s = 2\pi f_s$ and $\Omega_{\max} = 2\pi f_{\max}$.

Example

Human hearing: $f_{\max} \approx 20$ kHz

CD audio: $f_s = 44.1$ kHz

$44.1 > 2 \times 20 = 40$ ✓

Example

Consider the analog signal

$$x_a(t) = 3 \cos 100\pi t$$

- (a) Determine the minimum sampling rate required to avoid aliasing.
- (b) Suppose that the signal is sampled at the rate $F_s = 200$ Hz. What is the discrete-time signal obtained after sampling?
- (c) Suppose that the signal is sampled at the rate $F_s = 75$ Hz. What is the discrete-time signal obtained after sampling?
- (d) What is the frequency $0 < F < \frac{F_s}{2}$ of a sinusoid that yields samples identical to those obtained in part (c)?

(a) Minimum Sampling Rate

Given the analog signal:

$$x_a(t) = 3 \cos(100\pi t)$$

Compare with:

$$x_a(t) = 3 \cos(2\pi Ft)$$

$$2\pi F = 100\pi \Rightarrow F = 50 \text{ Hz}$$

Nyquist criterion:

$$F_s \geq 2F = 100 \text{ Hz}$$

$$F_s^{\min} = 100 \text{ Hz}$$

(b) Sampling at $F_s = 200$ Hz

Discrete-time signal:

$$x[n] = x_a(nT_s) = 3 \cos \left(100\pi \cdot \frac{n}{200} \right)$$

$$x[n] = 3 \cos \left(\frac{\pi}{2} n \right)$$

$$x[n] = 3 \cos \left(\frac{\pi}{2} n \right)$$

Observation: No aliasing since $F_s > 2F$.

(c) Sampling at $F_s = 75$ Hz

$$x[n] = 3 \cos \left(100\pi \cdot \frac{n}{75} \right) = 3 \cos \left(\frac{4\pi}{3} n \right)$$

Reduce frequency to principal range:

$$\frac{4\pi}{3} - 2\pi = -\frac{2\pi}{3}$$

$$x[n] = 3 \cos \left(-\frac{2\pi}{3} n \right) = 3 \cos \left(\frac{2\pi}{3} n \right)$$

$$x[n] = 3 \cos \left(\frac{2\pi}{3} n \right)$$

Observation: Aliasing occurs since $F_s < 2F$.

(d) Aliased Frequency)

From part (c), the discrete-time frequency is:

$$\omega = \frac{2\pi}{3}$$

Relation between analog and discrete frequency:

$$\omega = 2\pi \frac{F}{F_s}$$

Solve for F :

$$F = \frac{\omega}{2\pi} F_s = \frac{2\pi/3}{2\pi} \times 75 = 25 \text{ Hz}$$

Interpretation:

- Original signal frequency = 50 Hz
- Sampling rate = 75 Hz (below Nyquist)
- The sampled signal appears as a 25 Hz signal

(d) Aliased Frequency

Aliasing concept:

$$F_{\text{alias}} = |F - kF_s|$$

$$|50 - 75| = 25 \text{ Hz}$$

Conclusion:

- 50 Hz and 25 Hz produce identical samples at $F_s = 75 \text{ Hz}$
- Hence, 50 Hz is an alias of 25 Hz

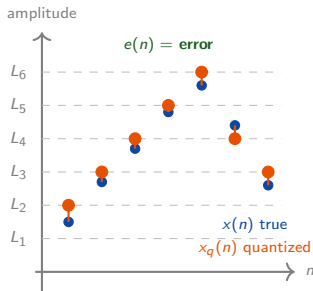
$$F = 25 \text{ Hz}$$

Definition

Map each sample $x(n)$ (which can be *any* real value) to the **nearest** value from a *finite* set of levels.

$$x_q(n) = Q[x(n)]$$

- With b bits $\Rightarrow 2^b$ quantization levels
- **Quantization error:** $e(n) = x(n) - x_q(n)$
- Quantization is **irreversible** (non-invertible)
- More bits $\Rightarrow$ smaller error $\Rightarrow$ higher cost



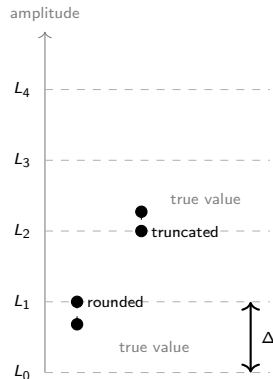
Common bit depths

8-bit	256 levels (telephone)
16-bit	65536 levels (audio CD)
24-bit	16.7M levels (studio)

Quantization — Rounding to Levels

Problem: After sampling, amplitudes are still continuous — infinitely many possible values. Computers can only store a *finite* set of values.

Solution: Divide the amplitude range into L equally spaced levels and round each sample to the nearest level.



Step Size (Resolution) Δ

Formula

$$\Delta = \frac{x_{\max} - x_{\min}}{L - 1}$$

where L = number of quantization levels.

Example: $L = 10$, range $[-5, 5]$

$$\Delta = \frac{5 - (-5)}{10 - 1} = \frac{10}{9} \approx 1.11$$

Levels are at: $-5, -3.89, -2.78, \dots, 5$

Key Insight

$\uparrow L \Rightarrow \downarrow \Delta$
smaller step $\Rightarrow$ higher accuracy
 $\Rightarrow$ less info lost

But more levels = more bits needed!

Normalization Before Quantization

Why normalize? We often want to map a signal from its original range $[a, b]$ to a standard range $[c, d]$ (e.g., $[0, 1]$ or $[-1, 1]$) before quantizing.

Normalization Formula

$$x_n = \frac{x - a}{b - a} \times (d - c) + c$$

Example: Map $x = 3$ from $[-5, 5]$ to $[0, 1]$

$$a = -5, \quad b = 5, \quad c = 0, \quad d = 1$$

$$x_n = \frac{3 - (-5)}{5 - (-5)} \times (1 - 0) + 0 = \frac{8}{10} = \mathbf{0.8}$$

So $x = 3$ maps to $x_n = 0.8$ in the normalized range.

Quantization Error

Definition

The quantization error $e[n]$ at sample n is:

$$e[n] = x_q[n] - x[n] \quad (1)$$

where $x[n]$ is the true value and $x_q[n]$ is the quantized value.

Properties of quantization error (uniform mid-tread quantizer):

- Bounded: $-\frac{\Delta}{2} \leq e[n] < \frac{\Delta}{2}$
- Modelled as **uniformly distributed** white noise with zero mean over $[-\frac{\Delta}{2}, \frac{\Delta}{2}]$
- Power (variance) of the quantization noise:

$$\sigma_e^2 = \frac{\Delta^2}{12} \quad (2)$$

Signal-to-Quantization-Noise Ratio (SQNR)

Signal power of a full-scale sinusoid $x(t) = V_{max} \sin(2\pi ft)$:

$$P_s = \frac{V_{max}^2}{2} \quad (3)$$

Quantization noise power (with $\Delta = \frac{2V_{max}}{2^b}$):

$$P_e = \frac{\Delta^2}{12} = \frac{1}{12} \left(\frac{2V_{max}}{2^b} \right)^2 = \frac{V_{max}^2}{3 \cdot 2^{2b-2}} \quad (4)$$

Therefore:

$$\text{SQNR} = \frac{P_s}{P_e} = \frac{V_{max}^2/2}{V_{max}^2/(3 \cdot 2^{2b-2})} = \frac{3}{2} \cdot 2^{2b} \quad (5)$$

Converting to **decibels**:

$$\text{SQNR}_{\text{dB}} = 10 \log_{10} \left(\frac{3}{2} \cdot 2^{2b} \right) \approx 6.02 b + 1.76 \text{ dB} \quad (6)$$

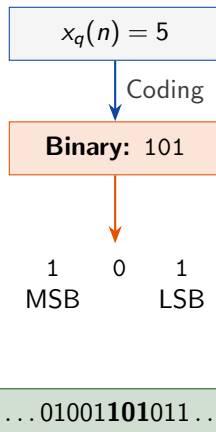
Definition

Represent each quantized value $x_q(n)$ as a **b -bit binary sequence**.

- Assigns a unique binary code to each quantization level
- With b bits: codes range from $\underbrace{00 \dots 0}_b$ to

$\underbrace{11 \dots 1}_b$

- Output is the stream $01011 \dots$ fed to a digital processor



Example

Example: 3-bit coding (8 levels)

Level	Binary code
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Coding — Turning Levels into Bits

Each of the L quantization levels gets a unique **binary codeword**.

Bits per Sample

$$B = \lceil \log_2 L \rceil \quad (\text{ceiling})$$

Example: $L = 10$

$$\log_2(10) = 3.32 \Rightarrow B = 4 \text{ bits}$$

Bit Rate

$$R = B \times F_s \quad (\text{bits/second})$$

Level	Value	4-bit Code
0	$x_{\min}$	0000
1		0001
2		0010
3		0011
4		0100
5		0101
6		0110
7		0111
8		1000
9		1001
		$x_{\max}$

Levels 10–15 are unused overhead

Complete ADC example: voice signal

- Signal range: $x_{\min} = -5\text{ V}$, $x_{\max} = +5\text{ V}$
- Number of levels: $L = 10$
- Sampling frequency: $F_s = 8000\text{ Hz}$

Step 1 — Resolution:

$$\Delta = \frac{5 - (-5)}{10 - 1} = \frac{10}{9} \approx 1.11\text{ V}$$

Step 2 — Bits per sample:

$$B = \lceil \log_2(10) \rceil = \lceil 3.32 \rceil = 4\text{ bits}$$

Step 3 — Bit rate:

$$R = 4 \times 8000 = 32,000\text{ bps} = 32\text{ kbps}$$

Why D/A Conversion?

The Problem

After digital processing, a signal exists only as a sequence of numbers. Loudspeakers, displays, and actuators require *continuous* signals — we cannot “listen to numbers!”

The Goal

Reconstruct a smooth analog waveform $x_a(t)$ from discrete samples $x[n] = x_a(nT_s)$.

Fundamental Guarantee

The **Nyquist–Shannon Sampling Theorem**: if $f_s \geq 2f_{\max}$, the original band-limited signal can be **perfectly reconstructed**.

Interpolation Methods — “Connecting the Dots”

A D/A converter fills the gaps *between* samples. Four common strategies, from simplest to optimal:

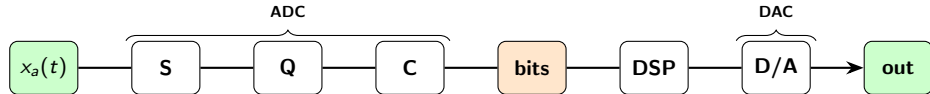
- ① **Zero-Order Hold (ZOH)** — hold each sample constant until the next; produces a staircase.
- ② **Linear Interpolation** — draw straight lines between consecutive samples.
- ③ **Quadratic Interpolation** — fit a parabola through every 3 adjacent samples.
- ④ **Ideal (sinc) Interpolation** — convolve samples with a sinc kernel; theoretically perfect for band-limited signals.

Method	Quality	Cost
ZOH	Low	Very low
Linear	Medium	Low
Quadratic	Good	Medium
Sinc (ideal)	Perfect	High

Practical note

Real DACs approximate sinc interpolation with a low-pass *reconstruction filter* placed after a ZOH stage.

Summary — Full Conversion Chain



Key Properties

- Sampling: **reversible** (if $f_s \geq 2f_{\max}$)
- Quantization: **irreversible** — always introduces distortion
- More bits $\Rightarrow$ less distortion, higher cost

Important Formulae

$$f_s \geq 2f_{\max} \quad (\text{Nyquist})$$

$$e(n) = x(n) - x_q(n) \quad (\text{Q. error})$$

$$\text{Levels} = 2^b \quad (b = \# \text{ bits})$$

Sampling

$$x(n) = x_a(nT)$$

$$F_s = \frac{1}{T}$$

$$f = \frac{F}{F_s} \quad (\text{digital freq.})$$

Quantization

$$\Delta = \frac{x_{\max} - x_{\min}}{L - 1}$$

$$x_n = \frac{x - a}{b - a}(d - c) + c$$

$$e(n) = x_q(n) - x(n)$$

$$\sigma_e^2 = \frac{\Delta^2}{12}$$

Coding

$$B = \lceil \log_2 L \rceil$$

$$R = B \times F_s$$

Key Relationships

- $\uparrow L \Rightarrow \downarrow \Delta \Rightarrow \downarrow \text{error}$
- $\uparrow L \Rightarrow \uparrow B \Rightarrow \uparrow \text{storage}$
- $\uparrow F_s \Rightarrow \text{better time resolution}$
- $\uparrow F_s \Rightarrow \uparrow R \Rightarrow \uparrow \text{data rate}$